

ActInf Livestream #021.02 ~ John Boik

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okay hello everyone welcome to the active inference lab to an active inference live stream with john boyk this is going to be great it's our second of these interactive discussions on the sequence of papers that john has written and today we're gonna be hearing about motivation and strategy as well as active inference of course a lot so i'm daniel friedman i'm a postdoc in california and i'm gonna pass it to john to introduce myself lead us into this second discussion while i share the jamboard link with our live participants so john thanks again for coming on and please take it away all right daniel thanks for having me on again this is great um all right so my name is john boyk uh i'm a courtesy faculty at oregon state university environmental sciences graduate program and uh we're here in this series of talks this is the second in the series to talk about the the um series of three papers that i recently published in the journal sustainability the topic is societal transformation in particular the de novo development of new societal systems new economic systems new governance systems legal systems educational systems and so on and last week as maybe you can see on the jamboard now last week we started with the first part which was on world worldview the topic the title is science-driven societal transformation part one worldview and uh we didn't actually get through that so we're going to pick up a few tail end ideas there and then hopefully start on the second paper part two which is uh subtitled motivation and strategy and the links for the three papers are there and i of course encourage people to download the papers they're they're open access so everyone can get them for free um page two perhaps yep okay uh so uh just for more information about the the project about the for the links for the papers for other links there's a bunch of uh little articles that you could look at if you're interested there's also an interactive model you could go to the principled societies project website and the address is given there so further information if you need it you can find it there cool three all right three okay all right so we uh we didn't get through the first uh through the paper all the way in the in the first our first talk so there's a few things that i'd like to catch up on uh before we move on to paper part two um we we touched on active inference uh in the in the lat in our last talk but we didn't really get around to discussing it so i want to say a few words about active inference itself a

little bit of a description about it for those who are not familiar um and then we have the topics of cooperation and communication and uh the the concept that complex organisms have complex needs and then on to some practical implications and this is really the main thing i wanted to finish up with is that in in part one i i provide some kind of high level sort of [Music] abstract concepts and i would imagine that the reader in certain places might be wondering why are we discussing this what does this have to do with actually building new societal systems so i want to talk about a little bit of the practical implications of all of this kind of abstract all these abstract notions that we've been working with so i would like to just recall just a few ideas from our last talk uh and one is the and that i i claim at least that is society has an intrinsic purpose and uh that and and by intrinsic purpose i really mean a biological purpose that we have gained through eons of evolution and that intrinsic purpose is the same purpose that every living organism has so every living organism has this purpose whether that's a bacteria or a human and then i also claim that society is an organism that can be seen as an organism can be viewed as an organism and a society has the same intrinsic purpose so if we're if our task is to develop new societal systems it really helps to have some ideas of well who are we and who are we building this for and what is the purpose of uh if we were to build a new societal system new economic system for example what what would be the purpose and if we know the purpose then we can take the remaining the next step and say okay we know what an economic system is supposed to do we know what a government system is supposed to do so is this new design design a design bead is it fit for purpose that's where that's the concept is that if we're going to design new societal systems de novo from scratch then we ought to have an idea of what they are supposed to do and then how we might measure their fitness so that we can determine did we did we create something that is fit for purpose and which one of these options which one of these designs might be more fit for purpose than others all right so that's that's kind of where we're going here and uh so the the i claim that the intrinsic purpose of a society and organism any organism or any organism is to achieve and maintain vitality which is accomplished through largely through cognition and cooperation and so to maintain and achieve vitality for who and that who i claim is the extended self so the extent this all of us are are enmeshed in our world enmeshed in our environment enmeshed in in a society with each other and if we're speaking of a body then the is those interactions with those necessary and vital interactions with the rest of the world so the extended self then is the self that extends to other people to to societies to other societies to the environment the local environment the global environment the biosphere so then in that

sense then if our purpose is to achieve and maintain fit vitality our achieve achieve and maintain vitality for that larger extended self that really goes all the way to the biosphere where all of life is in this together all of life is connected in one and uh and all of life is doing this thing doing this thing of acting to achieve and maintain vitality and over eons of evolution we get this amazing world of life and whales and sequoia trees and ecosystems and all these tea trees talking to a fungi in the ground and plants talking to each other and beads doing their thing with flowers and each other and i mean it's just like amazing coordination and communication and cooperation happening between species and within a species and within a human body like all of our cells so we just if i would say that if we're going to create a better society or create new systems that lead to a better society then the task will be to to fulfill our intrinsic purpose which in in a way means to simply get with the program that is get with the program of what all of life is doing around us right just align with what life is doing and uh then we will you know we'd be off to a good start if we can do that all right so that's uh that's the purpose and then i at the bottom of that slide i just i highlight the word wisdom because it is reasonable you know we we think of a wise action is something that is benefit you know widely beneficial and action that is aware of the needs of others and is as useful action but that is what we're talking about we're talking about uh of of taking actions that help to fulfill our intrinsic purpose and really by definition i would say that would be wise actions so in that sense we're aiming at a wisdom society a society where wisdom is valued and discussed and sought after and you know and and hopefully one in which uh individuals and groups and others institutions that act with wisdom will be recognized you know as such okay so i do ask the question there at the bottom can wisdom be reversely defined now often it's you know it's it's a term that is just generically applied to like to uh to uh you know actions that are that are some people feel are useful or appear to be useful but but if we're going to build new societal systems and we're going to have some sense of fitness for these systems and the systems are in some way supposed to help us to act with wisdom and to facilitate action with wisdom then perhaps there is some way actually rigorously define wisdom or to know when when actions are wise and i would say then that leads us right into the concept of active inference because because uh through active inference we what we might say that actions that reduce uncertainty and maintain uh you know reduce expected uncertainty about our are thriving or survival or survival into the future those are though those are benef those are beneficial actions and then again by definition they're wise actions so there might be some way to actually talk about wisdom in the same sentence as fitness right the the two cons two concepts concepts must be

closely related okay john can i ask i have uh two things i wrote down so one was cognition and when we talk about multi-scale cognition especially with a computational or a modeling bent it's easy to get caught up with cognition being a really mechanical process but i liked how you brought it to wisdom and so it's like if we're gonna be a cognitive system or if that's a framework we're taking are we gonna be calculators and just do zero through nine calculations or will there be wisdom in the system and that kind of brings me to the question which is clarifying wisdom is a big opportunity but also there are risks and uncertainties so here you know in 2021 looking forward potentially to some type of wisdom transformation what can we cling to and how can we follow wisdom because it sounds like a great idea but then how will we know it in the midst right right okay so uh your i'll talk to your first point uh first that excellent so so i i want emphasize that we're not talking about building a society of robotic you know linear thinking logical organisms you know that's not really what humans are anyway that's what no organisms are you know every organism has uh you know some degree of complexity of of its of itself and uh humans are very complex it will talk about the complexity of human needs in a moment but this it's life is not all about logic and uh it is partly about uh you know a good share of it is about heart and intuition and all the other emotion and all the other aspects that come to play in in what it means to be human so i'll repeat something i had said earlier too is uh you know the if we are to build better societal systems better societies those would be societies that would help us be ourselves more help us be true to ourselves more true to our hearts more true to our deepest longings more so i'm certainly not talking about uh building a society of legislation people doing logic or computational logic or something like that that's not at all what this is about this is about being more human more ourselves more capable of of uh making wise decisions right yep and and i think some of these ideas from the you know from over that that have been developed say in the last 10 years of science ideas from complex systems science ideas from cognitive science ideas from active inference ideas from evolutionary biology and some others are we're in a much better position today to be talking about you know how how might we do this how might we build systems that actually recognize the full human the full society the full the full interactions and then and then help provide guidance of of how how that might best be done right yeah just one quick point on that is people have talked about general systems theory and about how we could simulate or choose utilitarian outcomes or all these different ideas that have been percolating for truly thousands of years in many cultures what's different well in your pocket there's something that can simulate not the universe but it can do calculations that

complement human abilities in a really potentially beneficial way that's different it's on a continuum with other extended cognitive elements like books and communications telecommunications media and things like that but we're in a computational and a conceptual phase that's totally novel and so we need to consider um classical critiques like it can't be done it won't be done amidst the truly ongoing changes that's right actually exponentially uh expanding capacities for computation but your point is very well taken this again i do not mean that we're going to have computers deciding our every action it's not at all what i'm talking about i'm talking about becoming more human and using the tools that we have available to extend our computation and help us to understand better with the you know the complexities of the world around us right so uh right so this is this is uh humans taking advantage of all the tools they have to become more themselves and better and wiser and and better decision making better decision makers okay so uh then so that's what that's what we're gonna cover today uh this first part and then we'll move on to part two all right next slide all right four four so um so this is a slide on uh active inference and i just want to put again for those who maybe are not too familiar with the concept i want to give a kind of a brief overview um excellent uh carl friskin obviously is one of the you know the main forces behind this concept he's been working on it for well over a decade now uh uh there's a little image there on the left of the slide that i'll just say a few words about so we can talk two uh components to the universe uh the universe at large and then some individual that is in some sense separate from the universe right so uh the the concept of an individual is something that is that can be demarcated from the rest of its background so for the moment we'll say there's we're going to draw a circle around the individual and then the rest of the world is going to be outside that circle so the rest of the universe has hidden states it has a you could say it has kind of a in a sense a generative model of how it works how it functions uh when it is when it has you know is is acted upon by an individual it responds in some way it has in a sense it has a model to do that and the individual has a model of what the individual how it understands the larger picture now obviously the universe is bigger than a human mind so there is no way that a human mind is ever going to understand every little detail about the universe that's impossible so um a human then or any really any individual is faced with this problem of trying to act in a universe that is far more complex far more complicated and dangerous at times so how do you how do you act under uncertainty that's the problem of humans and other organisms how do you make good decisions under uncertainty and that's certainly the problem for society too how do you make good decisions wise decisions under uncertainty so a uh

when we talk about separating the individual from the rest of the universe we're talking statistically we're talking about independence from the rest of the universe and in this line of thinking that independence is granted by what's called a markov blanket so the individual knows its own beliefs or it has its own beliefs it can choose actions and it can receive sensations from the world so it can act and it can observe the the result of its actions and then the universe on the other half the universe is impacted by actions and then changes in some way has some dynamics and then that leads to sensations for the individual and then the individual receives a sensation so we have kind of a circular pattern going on here where the uh the individual acts well maybe we'll say this way the individual receives sensations after some action receive sensations from the universe and then it can do really one of two things it can it can it can alter its beliefs about the world maybe maybe the sensations tell the individual that it was wrong in some ways so maybe one option then is for the the individual to update its beliefs about the world that's uh new here or it can take some kind of action to better fit the work better fit his experience to to how it understands the world so if i'm cold if my sensation is being pulled my action might be to put on a coat so that don't freeze to death right i can bring the world a like try to align the world with what i expect the world to to what i expect of myself and what i expect of my own condition and how i expect the world to act so that's the kind of the two options we have available we can update our understanding and beliefs about the world and we can choose actions uh so that's really the gist of what that picture is and then over on the left on the right hand side i have a few equations that sort of puts that into kind of a formulative way so the first one that was just that's that's as just a slide taken from wikipedia but the concept is that because i'm an individual and because i don't i can't actually know what the generative process of the universe is i can only know my sensations and and my actions i would like to reduce the surprise that appears like i don't you know it's surprising to me if i'm cold and sick and hungry and harmed right i don't want those kinds of surprises i want some certainty about me existing into the future but the trouble is i don't know what the world's model is i have to essentially create my own version of it that's the the density q in this model i'm going to create a my best guess as to how the universe works and uh the universe has the density p in this equation and uh i'm going to do something actually that is it's not really common in data science so if any data scientists are listening there you ought to be probably very familiar with these concepts already this is a variational inference so it's a technique widely used in data science to solve problems and it and the id that's the central idea is you you have some complicated density p and you don't really know how to deal with it so

you do you instead choose a much simpler density q that is more better behaved and then you work with that and the whole idea is if if if you bring q closer and closer to p then you'll be closer and closer to actually having a model of the real model of the world so uh just in that term for the elbow the evidence lower bound is it's called in data science there's a kl terminal divergence term and that just simply means that term will go to zero as p and q come closer and the other term there uh $\log \log p$ uh that is really just the negative that is called surprise so uh um you the agent i would like to reduce the surprise that it it uh recognizes experiences and uh and doing and one way to do that is to uh is to also bring the the densities p and q closer so you get a good model of the universe and you act and act in a in a way that is consistent with your understanding of the universe and you update your beliefs as necessary to be have a better model of the universe and if you do all of that the your about how the universe acts and you're surprised about maintaining yourself into the future will be will be reduced so uh that's that that functional is called a free energy functional and the whole concept is an active inference is to use this action updating uh observing from the universe what the sensations are and then going in a circle and and updating an action and updating an action of you know updating action observation updating action observation uh the the the the idea is to reduce this functional of free energy and that fristen i would say claims that this is what all organisms are doing this is what life is doing and if this doing so leads to really all the structural diversity that we that we are aware of in our universe the ecologies and ecosystems and organisms and cooperating cells within a human body and all of that is is uh has has has become what it has become through a process through this free energy minimization process so uh so in a sense if we wanted to somehow uh talk rigorously about wisdom it would be in some sense at least you know from this perspective it would be wise actions and and learning activities and explorations wisdom is the kind of equivalent to reducing expected free energy i just want to make a just a few other comments here is this this is the simple this page is the simple version so uh what we really want to do is is reduce expected free energy like what will happen you know a tiger is running at me i could run i could fight i could jump on a car i could jump in a car i could do all these things i have these options paths these paths i could take these decisions i could take which of them is going to be best for me in the long run or in the short run and so that requires expectations of free energy over possible decisions of the in the future and that becomes quite complicated none of that is really in this slide but i do want to you want to everyone to realize that that's there john i want to lock in two really key points that you touched on here and then ask a question so the first is the first thing you said on this

slide is active inference is about the reduction of uncertainty so that's in direct contrast with the human as an economic maximizer or an optimizer of the reward or that we're trying to do some sort of optimization maximization on an outcome it's a process-based reduction of uncertainty that active inference highlights and so of course you're highlighting the outcomes that we want to see those are our values our preferences and our predictions we hope but we're going about it with a process based and uncertainty reduction based lens rather than simply like number is going to go up we got to get there so that's the richness of just if nothing else all the critiques of active immigrants aside to focus on the process and the reduction of uncertainty is really powerful and the second lock in is you immediately went from the reduction of uncertainty to collective behavior and it's like there's things that the brain knows that a neuron doesn't like how to speak english and there's things the colony knows and does that the ant nest mate doesn't so part of this is about realizing well there's things surely society knows with all of our diversity and all of our different skills that an individual doesn't so what does it look like to take that seriously that the cognitive systems are not just nested internally like well the brain the brain region the neural circuit the neuron the organelle yeah you can go down but you can go out and it's not just a cope and so what that looks like is a really important point so the two things there reduction of uncertainty with active inference and multi-scale cognitive systems and it's really important points um good or can i ask the question no excellent excellent yeah go ahead i'm just trying to catch up because you say a lot of awesome points in each of these breakdowns and the question is really um what kinds this is on the jamboard what kinds of social processes or data are promising for active inference modeling let's just say now maybe in the future we'll license other deployments but right now like what kind of data would you be curious to collect or find out about we may talk about it later in the economic or social context but what kind of data types are we thinking about making active inference models of yeah okay so to answer that question um uh i'm i'm going to ask that question in about two more slides so i'm going to ask that very question to you and to the viewers out there but i'll do i can give you just one that's simple okay we can come back to it but go for an example yeah i'll just i'll give one example uh because i also want to emphasize something else that you also just very briefly touched on so uh also i had touched on but i want to emphasize that if we're talking about expected uncertainty that implies that we are anticipatory so i just want to underline that every organism is anticipatory every organism is doing this reducing free energy you know i mean from this view and that requires anticipation of what the future will be and so so anticipation or you might call it forecasting

forecasting is central to um our our well-being our our continued existence i'm forecasting that if i see a tiger coming towards me that it's i'm gonna be it's lunch if i don't do something so uh you know that's i if we don't we don't think clearly about what's to happen next then cognition is there is no hardly any cognition just one one short comment on that is anticipation and anticipatory systems has been discussed in many other contexts like cybernetics and active inference helps us walk from this very simple beginning you said okay there's an inside and outside if you don't have that you know you need to think a little bit more clearly about the system or something like that but we can start with the simplest and then we can build it out to the sophisticated affect of inference with the affective charge and all these other modern models but we're starting with this loop which is why we always come back to these kinds of figures and why it's really important that you're making it at this level right good uh so uh just so so we can ask the question uh all right you know tell me like kinds of things what does this want to be good at forecasting uh and you know actually i'll ask this question a few slides so we don't have to go too far into it now but let's just just just take say governance uh or leaders political leaders if they if a political leader unders is good at understanding what's coming next what the future is going to be then great then the political leader can say hey we better do something because if we don't this thing is going to happen right but as a society today do we actually remember those predictions do we in any way like you know score accuracy of people's predictions or the predictions of science or the predictions of organizations is there like how does society today remember remember that predictions have been made and now the future is has evolved and come about and how who was good at who was going to anticipate in the future who who excels what what what what organization or political group or whatever who's been really good at understanding what's going to come next right so that's just i'll just leave that an open question because that you know it gets the mind working out that you can think of all kinds of ways that we don't do a good job of of remembering and anticipating today just to ask a question from the chat um someone wrote does john use active mathematical framework in his work for merely descriptive purposes so potentially as opposed to generative or like real-time decision making so where does where are you right now or where do you want to be with respect to using active inference to describe simulate make decision support etc right right yeah a good question so and maybe maybe the part of the answer to that question is to is to tell you what stage this project is at so the the series has just been published this uh the series proposes a r d program uh that we that that we that society could undertake and uh it is conceptual at the moment uh the the whole program is conceptual

it's not been funded it's it's i mean you're in a sense you're looking at it this is it it's the three papers and our conversation and whatever you know other people have to add to this so uh this is where we're at we're we're obviously focusing or i'm focusing now on the the kind of the the big picture conceptual ideas because that has to it's that has to happen before you actually get into the nuts and bolts of okay how do we use this what how does this go what kind of models can we build but uh i would guess and i would expect that if this project moves forward then at some point we'll be talking about the numerous ways that the society uses information and how it remembers and how it anticipates and what models might be used for different kinds of intuition participation and um and you know how do the nuts and bolts of that go together so that's to come that's the research project to come and i hope i hope we get a chance to go there that would be fantastic we'll lay it out as likely and then we'll make it happen okay there we go there we go all right so uh so that's that slides go on to the next sounds good five yeah and and maybe two you know before we move off the last slide i i do want to say that active inference has this wonderful uh part of it that i'm very attracted to and that is that it provides a way for an organism acting in its environment to play it by ear to to to learn as it goes to be to have a of learning a goal for learning an action that can evolve over time right so so so if we were gonna program an active inference robot we would not be giving it a formula for you know do this 10 points do that five points now find a way to act in the world where you where you you know maximize the number of points you can get it's not really like that and the world is for for in a real setting the world is far too complex for someone to pre-program how many points you get for certain activities or reaching certain goals right so a intelligent agent has to be able to alter its its understanding of the world and its goals at the same time to and we'll talk about this again but to balance exploitation of what it knows already to use what it knows about the current situation and exploration jumping into new uh new unknowns uh considering new proposals concerning new rearrangements so so the so active inference and really structural development and evolution of all of life and ecosystems and cells and communications between organisms and all of that is is kind of built on this balance between exploitation and exploration and you might think of that in various ways between old and new um you could even think of that in terms of like in modern day society to a degree you could even think of it in terms of liberal political versus conservative political in their best sense that is a you know conservative political might be more more interested in relying on tradition and then exploiting their current knowledge and perhaps a liberal might be more interested in exploring new ideas and and taking the leaps of you know into the into the unknown to to

learn something new and do and do something better to improve a process to change a process right so decision making is very much a balance between those two uh exploitation exploration and active inference provides a mechanism by which that can occur quite naturally i won't even go into detail to say that an action model and reducing uncertainty on that helps you act in unknown unknowns i think it's something unknown unknowns that's correct yeah of which there are plenty all right now slide five uh i guess this is following up on active inference just a bit um and i think we've talked on most of these things uh the uh yeah i think we have great the whole the whole point is to resolve expected uncertainty great so okay good next six all right so i just want to touch a few just a few ideas on cooperation and communication so uh actually cooperation and communication are you know go hand in hand right they may have even evolved together possibly so um cooperation is the is it really is almost the rule in the in the in our world life cooperates in so many ways so many ways and the more we study it the more we realize how deep the cooperative cooperation is with you know within a species within a within between species between trees and fungus between within ecosystems between ecosystems i mean it's just cooperation is everywhere and as i said last in our last show if you want evidence of it that is if you want mind-blowing evidence and cooperation just think of your own body the trillion of cells that are cooperating to for you to be what you are right so um cooperation exists everywhere and it is nearly impossible without like you cannot cooperate with someone if you don't co-op communicate and the richer the communication the at least potentially the richer the um and obviously by cooperation i mean black communication i mean honest dishonest communication is hardly communication right so we're talking about honest exchanges between cells between people between societies um so it's interesting though that you know like to take your body for example you have all these trillion cells and they don't have to cooperate i mean there's you know they cannot cooperate and sometimes they don't cooperate with each other and they can act independently to some degree right so there's always throughout the universe there's this tension between what cooperation provides and what the individual you know what the individual might want to how it might want to behave if it were just left to its own to its own right so why would anything cooperate and and the the kind of the big picture answered that question is because it reduces expected uncertainty it's again the it reduces free energy so by cooperation a an organism can assure itself of survival it can make its survival easier it can make communication easier it can you know do have all these benefits so nevertheless every given moment is this there's this tenuous balance between you know do i continue to cooperate in

this in this sphere or or or not and who and who is and at what to what extent do i cooperate do i cooperate within my uh family and or in my community and in and or in my society and or between societies that like at what level am i interested in cooperating right and the you know the answer the general answer that question is i am interested in cooperating on every level in which uncertain my uncertainty is reduced my my vitality is insured and my uncertainty is reduced right so it crossed the layers across scales but we do see this examples of when cooperation fails and one example that i like to use is the example of cancer in a body the cells cells transform cells start to not cooperate with the uh the uh you know with their milieu with their tissues with their with their environment so uh so my question for listeners is uh so we have an idea of what cooperation means and then what is the opposite of like you might think okay well the opposite of cooperation is selfishness and i would i would suggest you know maybe in certain circumstances that seems pretty clear but in a more general sense maybe that's not the opposite of cooperation right maybe maybe there's almost always cooperation and what changes is the extent of self of the agent or you know the organism that's that that is taking the action for example cancer cells cooperate with you know they cooperate with each other at least so um and they and they might cooperate within their you know within their very small tumor milieu there's a lot of communication that happens between cancer cells once a cancer forms a tremendous communication that goes on between cancer cells so maybe it's not maybe it's not so useful for us to think of cooperation versus selfishness but rather cooperation as a as a scale scales of cooperation in the extent like am i cooperating with my immediate family with my you know like or or or am i cooperating with societies that are far-flung and halfway across the world right to to the degree that we understand ourself to be extended we are cooperating with i mean you have to understand that right if i understand that i am part of this environment this ecology then my any actions i take to benefit the ecology are actions that i take to benefit myself right so if my sense of self is extended out to the world to the or to the degree that it's extended out to the world i am interested in cooperating with myself to better myself to reduce my overall uncertainty right so cancer perhaps is it's not really a matter of being selfish perhaps like i can't you know people say cancers are really selfish they're just gonna they're just gonna you know only pay attention to what's good for them yeah but that but they're cooperating they're cooperating with a smaller group and because they're cooperating with a smaller group and that because their extensive self is reduced their capacity for cognition is reduced you know we're talking we're just kind of anthropomorphic here but you know they they they would have a small tumor would have a more difficult time

understanding the long-term effects their long-term effects because their their cognitive data is to a small set of cells versus the human body which has the full cooperation of the brain and the feet and the hands and the understanding and the capacity to use your mouth to ask others what's going to happen next right so the more you the more you're the more you extend into the world and are cooperating with the world the larger your cognitive capacity the larger your problem solving capacity and the better you will be at understanding in the long term you know the outcomes of your actions and how that's going to affect your own um you know how it's going to affect you as an individual and you as an extended self okay so i got a little off there but uh you know what i'm just saying cooperation is everywhere and it's all communication and shared computational capacity and shared decision making the potential for more more cognitive capacity and better decision making great the only point i'd make here as a ant researcher and have thought about conflict and cooperation across scales is that it's really a story we're telling about a certain outcome like when the white blood cell eats a cancer cell is that altruism at the higher level is it a conflict at the lower level so it's a scale-dependent and it's often a perspectival narrative that we're telling about what cooperation is and um it's just like yin and yang there's a compliment between what we call cooperation and communication they're like the good guy and the bad guy they come into existence when we have fitness and free energy to dissipate so we need a story that allows us to act in that gray zone instead of getting fatalistic about competition or getting utopian about cooperation because this is not going to be necessarily a smooth and lovey-dovey ride with cooperation nowhere is it in the body so it's like a very nice nuanced way and that's why we fall back to active inference rather than an ism or a specific metaphor of choice like the selfish gene right excellent yeah and just to just to underline that this project is not about asking people to behave like super altruistically or something like that or to cooperate you know beyond what would be natural that that would never work it would fail it would fail because that's not what that's not how it work that's not how what life is doing that's not how life is working so you know we just really just need to be set up our system set up our processes set up our societal decision making you know ways we go about making decisions and just align ourselves with what we actually are and what life is actually doing and that means that there is a continual conflict between cooperation and uh you know and not cooperation between an and kind of an individualism and a more cooperative stance that's just like there's a ongoing conflict between exploitation and exploration between you know conservative and liberal between old information the value of old information and the value of new information this is a all of this is

a is a dance it's a dynamic dance right so but there is a sweet spot in that dance there is a place to hover a sphere to hover in where uh you're balancing all and new exploitation exploration uh conservative liberal you're balancing them and because you're on that in that sweet spot in that critical area you're able to do about the best that any organism could do under uncertainty and life the magnificence of life is the proof of that awesome two seven okay speaking of life yeah right so so uh in our last talk i had mentioned that um that complexity is uh you can think of it at least in a certain way you can think of it as problem solving capacity organisms complexify and environments complexify ecologies complexify in order to solve ever more difficult challenging problems so it is not true that that complexity is just a path to to chaos and failure right systems do fail i mean obviously that does happen there's there's complex systems that can fail but by and large complexity is a path to in greater problem-solving capacity so humans are complex so you know of all the organisms on the planet we're pretty complex we have complex brains we we have complex cognitive processes we can we think about what might happen in a thousand years and we're concerned about what might happen in a thousand years that's you know that's seen that's enormously complex say compared to a bacteria that doesn't have a central nervous system so because we're complex and this is true for all organisms in general that the more complex an organism is the more complex it's needs all right it's the more the more feelers it has out in the universe the more more extended it is into the universe in a sense and that is certainly true for humans and uh this this table is some needs according to maslow maslow and i just want to mention these as an example not not that these are the needs we should focus on but just as an example of of needs he i think he has maybe nine or eight or nine of their i think but affection we we just we want to be loved and we want to love both you know we want to share friendships it makes us happy to do that we we want to be creative we want to create things we want to build skills we want to have freedom to associate who with whom we want to associate and freedom to move about and look around and poke around and look under things we you know we want to participate we want to cooperate we we want to understand we want to learn learning is fun for you know hopefully learning it's fun we want the rest when we're tired we want to get some exercise we want to move about you know we need to relax sometimes like all this is these are if we can call these core human needs these speak to something that i've i think briefly mentioned the last uh talk which which is called essential variables so if an organism is if it's purpose intrinsic purpose is to maintain vitality into the future well you know what what are those variables that are important in maintaining

vitality you know like you can think you could think of a few important things right off the top of that like fresh water to drink you know clean air to breathe the food shelter you know like you could start to think of what some of these essential variables might be but i would like to suggest that some of these core human needs somehow fit into the concept of humans need to be humans need affection you we will be sick if we don't have affection right we will be sick if we don't have freedom we will be ill and we won't be able to and and and we won't be able to solve we won't be able to be good problem solvers in our environment if we're if we're hampered by a lack of meeting you know will be will be worse so the point of all that is when we think about a complex organism like a human we can think about complex needs and then we can think about okay so now it's starting to become clear when we're reducing uncertainty we're reducing uncertainty about these essential variables which might touch on things like the the eight or so nine uh core needs that i'm listing so i wanted for example i want to have i don't want to be uncertain that i will be loved in the future you know or that i'll have food to eat or that i'll be able to move about freely i would like to be fairly certain that that is i'm going to have those freedoms i'm going to have these needs met right and so again then we're not talking about some big computational calculator thing that for this project is this is this is about heart and soul and being a human even more human than society allows us to be today like you know society is not set up our society is not set up to to focus on providing for these complex human needs right i mean we have poverty we have billions of people in poverty in the world and the you that's already like all kinds of needs are not met because someone doesn't have the you know the financial capacity to to do what they need to do right so uh so uh we can we can there's one last thing on this slide this little phrase that i like to use is the solving problems that matter so society might solve all kinds of problems for example if i work for a manufacturer that is producing some like you know let's say a some commercial relatively useless product that people are uh that they're that people are buying because the commercialism is is pushed you know i made maybe i'm solving the immediate problem of making an income so that i can pay my rent but beyond that am i solving problems that matter you know i don't have a good example maybe someone else does the useless thing that many people do in their with their life's work you know their 40 hours a week is doing something useless there was a book there was a book there was a book about that a while ago and they had a phrase but i forget exactly what his phrase was well i don't know if it was a family-friendly phrase but i think i know it but yeah i don't think it was i hesitate to say it because it wasn't family friends friendly you know it's like you know uh whose team our team like that

kind of a call and response like which problems the ones that matter i mean then what does it mean to matter well that's our preferences and our affordances and then as far as cases where the work or the problem doesn't matter there's times where the person might think that the problem matters but when you pull back it's actually a non-linear response where even if it does matter the issue that they're concerned about that their effort may not be the most efficacious and this isn't going to go back to another optimization you must be engaged in the most efficacious um because we're actually putting space for play and identity and participation understanding all these attributes that aren't just some simple cybernetic like action selection but it's really important to keep this as our guiding light it's the problems those are the salient ones the ones that matter but we need to be informed by our values yeah yeah and obviously we're talking high level here we're we're trying to give a context for this whole thing of what could society be and how would we go about measuring fitness and you know what is a society supposed to accomplish and for whom and then we have to talk about the you know these these high level concepts because they can be guides for us right so the just the phrase solving problems that matter it can be a useful guide because it reminds us oh yeah what what matters and what am i doing with my life and what are we doing in society with our lives are we doing things are we solving problems that matter and you know obviously there's a whole bunch of enormous problems that we're not solving right like climate change and other things you if society was solving problems that matter that would be like enormous armies of biologists and you know other people the uh ecologists and field workers and all kinds of people you know focusing on the problems that matter most for for happiness and survival and currently in today's world you know that that is heating up heating up to be climate change and biodiversity loss and ecological destruction and pollution and groundwater depletion and like all the things that that were were pushed you know all the damage we've done to our to our certainly cleaning those problems up ought to be among the problems that matter okay so uh there we go i think uh uh one last thing on this slide photo or in the bottom right i i bring the topic of culture in and i just want to just say a word about culture from an active enforcement a viewpoint so you know at its best culture is a remembering of what is important to teach you know to to not only to teach new people to each new you know new young adults as they mature what's important but also to to offload some of the computational burden that individuals would otherwise face so if i have to figure out everything for myself from you know from scratch as an organism oh that's tough it is so much easier for my for me if my peers you know point me in the right direction tell me you

know give me some indication of what's dangerous what's not dangerous what's important what's not important give me some guidance i can learn so much faster if i have some good guidance and uh and also if i can just rely on culture to just to you know like okay you know my culture does this thing and i'm not gonna maybe even think about it too deeply because i'm you know like if my culture is good it's maybe reflecting in some capacity this importance of putting attention on those things that matter right so now we've now talked about two important things here actually uh culture and how culture relates to uh to the reduction of uncertainty overall and then uh also attention and the role of culture in directing attention so those are themes that we'll be coming back on nice so attention attention and essential variables you know like those go together right if we've not obtained attention to essential then our whole free energy minimization thing is not functioning correctly right because we have to put attention on those things that are important okay all right next all right so now we get to the slide where i get to ask you any any listeners like what does this mean practically so we've talked about all these high level concepts we've given some context for you know for who we are and what we might need and what a what a societal system is and those were all fairly high level disc discussions and now it may be it's time just to talk a little bit about what that might actually mean in a practical sense so the title of the slide is practical indications but up down at the bottom i have also known as how do you solve problems because the two are the same thing like this is the question how do you how do we solve how do we how do we um how do we how do we solve problems how do we focus on uh if we're good at problem solving if we're good at solving problems how do we do it and uh obviously we want if we're going to build new societal systems we want to build societal systems that help us that facilitate our problem-solving capacity facilitate our capacity to reduce uncertainty expected uncertainty right so i'm just going to list some things and maybe on the first you know in the first read of this just you can we can just acknowledge like okay yeah that's how we solve problems we number one there we get we gather data we we we sense our world you know obviously that's how an organism solves a problem and even knows that there's a problem to solve for that matter and we remember we remember what we did maybe last week or last year we remember how we solved something before we remember that this situation is dangerous and we might want to avoid it in the future so we remember remember we anticipate or make forecasts and obvi we've already talked about the importance of anticipation forever for cognition in any organism and and cooperate we cooperate with those around us we we have a problem to solve and we get some help maybe we solve it together we we work together we

build a house together we we you know build a jumbo chat together we do all these things together we farm together but what's uh so cooperation is one way to solve a problem and then what does cooperation entail well it requires that we maybe we as a group we reward uh pro-social or cooperative behavior and maybe penalize anti-social behavior you know i mean that's in game theory they've done all these studies about cooperation and it's pretty you know there's some really good ideas pretty clear ideas basic ideas of when do when how do you actually facilitate cooperation and and what makes cooperation go down the toilet right so uh if your communication really helps cooperation rewarding uh pro-social behavior transparency helps cooperation fairness helps coop like all these we know all these things right from from common experiences in life and also from studies in game theory in other fields so uh so solve a problem we cooperate we communicate we we direct attention there's that word attention again we direct our attention to those things that matter we learn from our actions and observations so we update our models of things when we realize like oh no i guess the world doesn't work that way i should i should learn something but the world is a little more it works differently than i was anticipating or i thought before uh and then we act to reduce expected uncertainty we experiment we we we have goal directed activities our actions where we you know try to that's the exploitation aspect and then we also have the exploration aspect action for epistemic and again then we have this exploration versus exploitation trade-off so so i'm not i'm just saying what i think everyone already understands i'm just making a list here but now the question is okay fine so that's how we solve problems how do each of these play out in a societal system setting so i you know if i think if we wanted to we won't because we don't have time but i think if we wanted to we could list like 500 ways this this any of these play out and it's maybe should play out or could play out in a in a healthy society is a society that's good at solving problems uh i had already mentioned one earlier so like let's just say on the topic of of anticipation wouldn't it be reasonable to somehow uh remember who remember and pay attention to who is who is good at anticipating the future so if you know if a certain political figure or uh you know a certain organization is that turns out to have a really good track record of accurately predicting what's going to happen next well then maybe we want to pay attention to them and pay more attention to them to uh than to others who have a terrible track record of of uh predicting what's going to happen next you know like if a politician says oh if you do this if you do a then b is going to happen well if you know if by and large those kinds of predictions are wrong then you know politicians probably not worth much but but not just in politics obviously we may we anticipate and like for example that's what science does

science is you know awash with trying to figure out how the world works and what's going to happen next and so so that's another area that we could you know a healthy society might really support the scientific uh you know the scientific endeavor and pay attention to what scientists are saying like for example with climate change and other things you know if the whole scientific world is saying hey this is serious you've got to pay you've got to do something about it well then maybe a healthy society would actually do something about it right um honesty the honesty i'm just picking out things off the list you know but i think we could talk for numerous hours about how any of these might play out in a in a real setting right um honesty so uh look at look at the dis look at the look the amount of dishonest uh information or manipulated information that is flowing around us today in this world um you know why is why is that i mean obviously a healthy society would want share uh honest and valid information let's call it quality information you know models are good models are built on quality information so you know why here's a question why in our world why is there so much just information and i would i would suggest that at least some of it is due to the fact that individuals and organizations corporations are fantastically rewarded for putting out dishonest information right i mean i mean just think of a thief i can i can call you up in a phone scam and if can convince you that i'm your bank and i need your account number then i can have enormous rewards for being you know dastardly dishonest you but but it's true i mean maybe that's an extreme example but it's true in just ordinary examples of business and life one can be rewarded greatly or being fairly dishonest or you know putting out something less than quality information and then you gotta then you gotta ask okay well then if if you know like we don't society doesn't want to reward anti-social anti uh you know and anti-transparent anti-honest uh behavior how how how would a society reward pro-social honest behavior transpl how would it maybe a society could become more transparent so transparency obviously to be a part of but but um i'm i'm sort of honing in i don't know if you if you can tell but i'm sort of honing in just on this little question about how do we what how do we reward behaviors in this world as a society and then that suddenly brings up the question of the how does an economic system work and how does what what how do we view money and how do we view income and how do we use money and income in our lives and is there some way that we could reinvent an economy and reinvent money so that it wasn't rewarding uh you know reinvent rules and reinvent whatever we need to reinvent so that we are actually day-to-day moment to moment we're rewarding cooperative pro-social behavior and and honest behavior so like how do we do that now i have some ideas on how to do that

but that's in the part three of the paper so we'll get to them later so i don't want to go much further on this but i'm just suggesting that that that we could spend you know each of these could raise a hundred ideas of how to what we could do in a healthier society to improve our capacity for group problem solving nice points good for nine yeah i always think i am but but hold on one second on the right hand side of slide eight are a few other ideas so um uh you know obviously we've already talked about fit the societal systems measuring the fitness of them so um you know how do we how do that's also a general answer to this question of how do we build a society that is more fit well we actually measure things that are important and we score them and we share that information widely so we know how we're doing so we know so we can evaluate either our systems or our organizations or our institutions or society at large or those kinds of things so metrics are important i already mentioned science and critical thinking are important so that touches on the educational system for example the healthy educational system would really value critical thinking and would train individuals to be critical thinkers at an early age i would think and then obviously then removing the barriers to cooperation this society has enormous barriers to i might want to get it for here's just one example out of what could be a million examples say that i was kind of uh wanted to make the world i wanted to use my life's energy my work time my 40 hours a week of work a week to make the world a better place so i want to jump in and i want to uh i don't know let's say i want to teach teach kids little children you know maybe i want to be a teacher be a good teacher well that's great fantastic but if i do that i have to pay a financial penalty like i'm not going to i could become a banker instead of suppose those are my two choices i could become a banker or a finance minute finance person or something or i could become a teacher well i'm going to society is going to reward me greatly for being a banker or a finance person and it's not reward me very well for being a teacher so i might want to cooperate i might work to for the betterment of humankind if i do there's a good chance i'm going to be relatively poor you know like i pay a heavy price for that and that's a that's just a barrier to cooperation cooperate we should design systems so that the cooperation becomes the default easy no that like the do it this way because it's good for everyone it's good for you and uh sure you want to teach well great we have come on in here's here's opportunities for you to teach and you're not going to pay a penalty for becoming a teacher you know you're this is going to be easy you can be if you want to be a teacher and help young children learn better great here's wonderful opportunities and you're not going to have to pay a penalty for doing that we're going actually just the opposite we're going to reward you for doing it okay now we're done with uh slide

nine or slide it sounds good to nine okay so um so i the only thing i really want to say about this this slide is that this the concept of the societal systems as a cognitive architecture i you know i you know probably somewhere in the literature there's something you know talking about that in some sense but i it's i think maybe i don't know if it's as kind of like pointed as i'm making it in this series and there and there really is no name for what is what is the name for the set of societal systems that helps the society that facilitates societal cognition what is the name of and there isn't a name you know it's it's not an economic system and it's not a governance system because that's just those are just some systems that are that have their own connotations so what is the name of this whole set of integrated systems that helps as society to think well there is no name so i i mean i there is no formal name and so you can't look at if you're going to search in the literature on this topic there's no there's no term to search for and uh so i thought maybe it would be useful if i try to suggest some terms at least and i and i suggest the term uh societal cognitive architecture for the general the general idea that societal systems are a cognitive architecture for society but then for the more specific developing de novo systems to be good at facilitating societal cognition i suggest the term sales uh societal societal active inference and learning systems um i mean maybe there's some other better trim i don't know but i i threw together this one so it's sales that i'm interested in developing new new designs of societal systems that excel facilitate uh sell facilitating societal so uh you know if one of these terms catch on maybe then in the future we'll be able to if we're interested in the topic we can actually do a search on google and we'll find a you know 100 papers or a thousand papers on some of these topics well just one quick on that it points to the need for improved cognitive architecture such as knowledge management tools that help us collaborate and find information semantically like information is for our experience because you're right it is fragmented and therefore unproductive across truly different disciplines and different jargons different languages as well unscanned documents things that are just only on someone's uh notepad just it's all over the place right and and and sometimes a thing doesn't really exist until you name it and then it and then it sort of exists for the larger it can't exist for the larger society so i'm name i'm naming this because maybe if we name it we can talk about it you know yes the other the other problem there is like when i go to expl it would be much easier for me to just throw out a term like cognitive social cognitive societal architecture and if people understood that term then i wouldn't have to spend a half hour trying to explain what it is that i'm talking about right so terms terms can help if we have a shared understanding of what their meaning is what what is not sales because it's a you know what is what's not part of

the function of the body how do we differentiate what is apart from that instead of this just being the one-to-one rapper around economics governance legal et cetera et cetera right right right good point and uh i don't i really might my use of the term sales i'm suggesting it for use really in this de novo design you know of of creating new systems so um quite true the the the the you know the cognitive the concepts of cognition that we've been talking about they are universal so uh it is true but we're kind of kind of focusing on just sales as a term for the kind of r d project r d projects to move forward that might uh look at the de novo designs of new systems right just the pond traveling sales person it's a problem you know it's a hard problem actually it's traveling salesperson problem yes it is all right so with that uh we are uh uh ready to move on to uh the actual talk cool today drum rolls please drum roll schedule the awesome the second is the second part you know this is um a different media we're kind of spilling above and beyond what you've wrote in the papers because the three papers are the version of reference that's where people will cite but actually in this increasingly digitized insighable world perhaps the things that you're sharing with us as well as even the anonymous content that people produce on a jam board or live chat like this all becomes part of how we produce our uncertainty and act yeah and and hopefully becomes part of the record and part of the useful record for for moving forward on some of these okay so part one was all about world view it was you know high level discussions high level topics about what creates a world view and how what kind of world view would be useful in the setting for developing new systems and then now we're moving on to the more nuts and bolts of things a little bit more in the sense of why are we even motivated to to transform and uh and if we do want to develop new systems uh what kind of strategy you know what like it's it's not enough just to come up with some great design it has to be viable in the sense that it can be implemented there has to be some way to actually use these ideas actually live these ideas um and if there's not then we're sort of wasting our time so strategy is core to the whole proposed r d program right um so yeah so i'm going to in in part two i talk about really uh i try to argue a given narrative in part two that i try to argue that transformation is prudent given the risks we face attractive given its potential benefits and achievable given the current political financial and social constraints so that's what i hope to accomplish in this paper and that's what i hope to accomplish in this this talk too cool all right so on the question of motivation like why transform we we touched on some of these ideas in our first talk but i i would just like to point out that we face a host of extremely difficult problems climate change being one of them climate change biodiversity laws groundwater depletion soils depletion poverty financial uh uncertainty uh you know in

that sense debt might play into that role um you know if you make a list of all very serious problems that humanity faces it's kind of a long list migration and then one problem causes another so climate change is certainly likely almost certainly will increase the number of refugees trying to cross borders people living without homes that are fleeing areas that are not that don't have enough water or are not livable or you know things like that so one problem causes the other and you know what what what everyone is fearful of happening is is that you have this the collection of 10 or 20 really serious problems and then something happens to flare one of them and then suddenly that problem you know transfers to other problems and then you have uh just one you know like everything's connected so if one part of our global society gives sway it could be that everything starts to kind of fall apart so there is a fear of that and there is a need to address these problems independently and then also in a holistic sense and i would like to emphasize that these problems we face are in a sense symptoms not really problems in and of themselves they're symptoms of a failure of society up you know to today a failure to address the needs of of the society a failure to reduce uncertainty and failure to do exactly what we were talking about in the previous part of this talk right so so climate change climate change is because we failed to take action we've known about climate change for 60 years i don't know i mean like it's it's been in the it's been it's been um uh in the scientific world like in a in a major way uh with major agreement on the implications and the basic processes and and what it means to society that's been talked about that there was you know the scientific world was on this 30 years ago and talking about it and writing papers and having you know so so we didn't society didn't listen and it didn't and i would say it didn't listen for a reason because if it had listened to those warnings it society would have had to have changed and there are elements of society that are doing well and are maybe not so interested in changing behaviors and especially changing power distributions right so uh the people that the power the powers that more or less guide we're not interested in dealing with our serious problems because we have they have gotten worse over the last 50 years or whatever these problems have amplified if you don't solve a problem it tends to fester and get larger that's exactly what's happened now we're at the period of not only not only will climate change you know is climate change going to affect the gdp that's that's that's uh yes it's the real problem is are we going to go extinct i mean this is serious this is you know like this is life and death now for if not if not millions or billions of people it's life and death for our species we have to deal with this in some kind of bold way or we we won't have children on this planet and you know whatever in uh 50 years 100 years 200 years humans won't be a thing so

that's how i say that because uh that's how badly society has failed us we are at the brink of falling off a cliff we maybe we're already maybe we already fell off a cliff and we just don't quite realize it yet like what what kind of healthy organism puts itself in commits suicide like what healthy system commits suicide none really only the phoenix yeah only the phoenix yeah and you know that maybe it worked out for that one okay it sort of it rose i guess but uh and maybe we'll do the same but uh you know my point is is that the serious problems we face are all symptoms of our lack of cognitive capacity at the societal level and there were there have been there are there have been solutions to all these problems for decades and we simply have not taken them because because doing so would have disrupted i think the distribution of power in society and that just people weren't willing to do okay so um so in this paper on this call for transformational change so i am i'm trying to i'm calling for transformational change right now but i'm not alone so this this idea of we need bold change this is not just me in a vacuum this is uh most of the scientific world really and uh most of the global policy making and a large section of the global public is all realizing that we are in deep trouble and we need to change not incrementally not slowly but rapidly and deeply and across every aspect of society so i'll just use the words this is a from the 2019 global assessment from the ipbes the un uh they as they in their words uh current and national they say current national global programs are insufficient to avoid you know catastrophe and they call for a transformational change they use the word transformational and they define it as a fundamental system-wide reorganization across technical technological economic social factors including paradigms goals and values i mean essentially uh you know enormous fundamental change in how we understand ourselves how we understand the world how we behave what we do for a living how industry works how consumerism works what products we make how we recycle how much energy we use how where we get our energy so so um i think maybe i don't know i don't want to put words in the human's mouth but i think maybe they were really still kind of talking about um you know like improving current systems you know like we can improve our economic system to do this we can improve our we can take bold steps within our governance systems we can take bold steps within our economic systems financial systems we can do better there's ways that we can do better i would like to very much agree with that there are ways we can do better we should we should i would i would put that under the category of reform we should do everything in our power to reform the existing systems because that is the quickest bang for the buck i mean we can do more isn't it imperative that we act and put our energies behind reforming as best we can the existing systems working within existing

systems and reforming them so they can do a better job is so that we can reach some kind of transformation even within existing systems but but this uh this proposed r d program and my interests are beyond that so i'm not i i'm not you know my interest here is uh not in reform not in improving capitalism for example or or representative democracy it is in building de novo systems from the ground up that are fit for purpose as as we've been talking about so you know i that's what i mean when i say societal transformation i mean i mean shifting to building testing and shifting to new and in the you know we're familiar with these kind of concepts when we download software and in the software world there the terms are usually updates so you know i'm going to update my software minor improvement i'm going to upgrade my software to a new version i'm going to get you know version four instead of version three so that's kind of like big reform i'm gonna get an upgrade then migration so i'm gonna do i don't even like this this company or this product i'm gonna switch to open source and i'm gonna use the linux operating system or some other i'm going to do something else i'm going to use some entirely new system because the old one really doesn't work very well so so my focus is on this the you know is a migration the the testing development uh uh and uh implementation of de novo designed societal systems um okay so i think uh you know and and a science driven this in a science-driven process so that's important to add to okay so lots of people calling for transformational change and that's what paper is about transformation understood as migration to new systems um on slide 12 uh i think maybe i mean others might disagree but i think maybe this is the first series in the literature that uh in the science literature that really lays out a r d program that aims to tackle this de novo design of new um i don't i've not seen any others that are that are kind of comparable so i think we're you know paving new ground here perhaps at least i hope uh i want to emphasize something we talked about on the in the last talk that this is uh we're talking about second order sciences as opposed to first order science so first order science again is what we normally think of as science as someone is doing experiments they're detached from the experiment the scientists are detached they're just they set it up they let the experiment run they they don't interfere they they just record what happened and hopefully they gain some knowledge from doing so fantastic stuff most of science is all this is first order science is one is wonderful but for these kinds of problems for this kind of problem that we're facing this global climate global crisis problem this i think uh requires what is called second order science and there's a in my series i i cite the work by phase you know on this topic and that's a this great article i i recommend that to everyone that it's in the paper so you can get it there but he describes what they i should say today it's a long

list of authors describe what they mean by a second-order science and why it's important and uh they just kind of in a nutshell we're talking about in our case building new systems we're talking about kind of a meta learning so we're learning to learn we're we're learning to build new systems that help us to learn and build new systems right we want to excel at learning that's the whole concept of this we we want a society that excels at learning and problem solving and adaptation and we're and we're in the process of learning to build new systems that produce this effect so it's kind of a meta learning science uh meta learning process and and just in general the second order science is the focus is on process the effort is value driven and the part and the scientists are participants along with stakeholders who together learn develop and evaluate solutions together so it's quite different from first order science and it's exactly the kind of science i think that is necessary for this larger meta learning project in the in the paper i i i i talk about six overarching systems i just want to say that that's what we're focused on here so economic uh their economic governance legal health analytical and educational systems those are the ones i've chosen to focus on and each of them can be split down further like in in my terminology the economic system also includes a monetary system and a financial system and so on and each of these can be split down further so that's what we're doing we're like trying to develop designs for each of these that are integrated not not just hey let's just build a new economic system and leave everything else and change we'll do the economic system first that's not the that's not the point of the process the idea is how can we build a integrated set of systems of full cognitive architecture that helps us to achieve these goals and it's really a nice dovetailing with development is what organisms do that's embryonic development or colony level development or the niche being modified so it's like research and development research information foraging um whatever we want to think about information foraging to mean here with research or just search but it's like we already have design latency within potentially even the structures we have so how are we going to pull out of the tailspin if that's what we're in to build towards developing a healthier trajectory that's right we're in the process of evolution you know and every organism is evolving exactly as you say and and developing so we're just being a part of that uh and in a you know quite a quite a sensible part of that i might add too it's kind of like oh we're faced with severe problems that we might uh have a masked eye off of humans or go extinct but because of so let me think what would be a good idea oh let's do let's think about doing things differently let's try to build a different way to live our lives you know that's just kind of like duh that's a good idea let's let's find a better way to live our lives and be more cooperative and have a

healthier world that's an excellent idea um okay so uh in this is kind of a warm-up to what's coming in the paper i i speak a lot about local community and i just want to emphasize or i want to explain what that means a little bit so um uh for one thing i i call society a organism or a superorganism and a cognitive organism with that and uh and by society i mean really a society of any size so in particular i'm interested in local communities for some reasons it'll become clear shortly so a local community might be just like a fraction of a city not even a full city and not even a political body not a like a formal political body but a group of people who in particular have in this case are particularly interested in uh participating in trying out and field testing and using some of these new systems that we're talking about developing so uh you know let's say uh when i say a local community let's say a group of a thousand people or more could be could be larger but not not necessarily even a city and certainly not a nation okay so um so i said that the u.n and others are this is slide 13. i've said that you the u.n and others are come out with strong language uh calling for a transformational change which i think again you know for many uh folks involved there i think maybe their idea really is more what i would call reform again very very important but it might help here to talk about why why reform might not be enough like why why you know if the fee win and other agencies and other groups are really are really putting up some great proposals for reform and improvement of existing you know why am i interested in in developing new systems why don't we just do reform you know if that'll do the trick let's why why bother with more and there's numerous reasons that i lay out in the in the paper number two of why i think transformation new system development is is necessary and uh uh you know to start with and as we've kind of been saying here i you can look look at the world and the problems that we face with climate change and biodiversity loss and others and you can suspect that current systems are incapable maybe incapable of preventing widespread catastrophe and securing our continued vitality i mean really the you know the the report card doesn't look good for existing systems they've done a rather crappy job of protecting the human health in the environment to the to the extent that we now face perhaps in our lifetime or soon after perhaps a mass die off of humans or even extinction of humans so you know why why would we think that they're capable now suddenly of of of acting differently you know excellent to do reform i don't want to talk anyone out of the reform efforts because they're absolutely important but i am suggesting that maybe we should have a plan b in case plan a doesn't work out so well so uh uh you know transformation this is this is a plan b in a sense and in the end might turn out to be better than plan a even if reform is useful and and helpful plan b might turn out to be the

better bet so you can kind of catch it on both ends there so i just talked about a few things i mentioned in the paper of why you know like why why why evidence of current systems being incapable of preventing a widespread catastrophe so one is the paris agreement and then this is quoting from one of the un studies i believe uh it says ambitions are broadly consistent with cost-effective pathways that result in a global warming of about three degrees by 2100 with warming continuing afterwards okay so you know in a practical sense you know if the paris agreement were interactive that's sort of what uh where it would lead to it is clearly it is you know is maybe an important agreement but uh not enough not nearly enough it looks like and uh if indeed we were to reach about three degrees centigrade change from um historical levels uh that is that's already in the iffy category of like you know how do humans what do we do we even have a civilization if we reach their degrees you know it's an iffy question maybe we do maybe we find a way to deal with it and get by or maybe we don't maybe that's already way too much um and uh also the reform efforts uh don't adequately address certain structural uh problems that we that we face like most reform efforts are not really looking at the extreme income and income and wealth divisions in the world uh you know the most of the most of the reform efforts are not looking at how we can not have billionaires but rather have everyone more or less equally engaged in the you know economic sphere and in a sense having economic power to make decisions in this so so even if reform efforts are useful and good i hope they are uh maybe they won't be enough and maybe they won't even touch deeply on some topics that really need to be addressed um so i argue you know this this section covers several pages but in the end i argue that even if even if reform efforts are are useful and helpful uh and they're successful and you know for what they're shooting for they're still likely to leave societies hampered in their in their capacity to learn and adapt with many unsolved problems to address like even for example even right now even at one what we're at i'm not sure what degree we're at right now over historical levels but still below two degrees maybe we're 1.3 i don't know what it is already there's nations that are you know they're the island nations that are not able to handle this you know like people are already started i have to relocate and every indication is that that's just going to become worse and worse so uh even if we even if we keep the keep it below 1.5 degrees or we keep it below 2 degrees the warming we're going to have severe problems social problems economic problems environmental problems ecological problems and somehow we're going to have to deal with this harsher world than even harsher than you know more dangerous than today and more strained than today more stressful than today and i'm suggesting that the best way to deal with a future

world that is stressful and uncertain highly uncertain and difficult is to become really good at problem solving isn't i mean that's how you would go about it that's you would be if you can you would become really good at acting in an uncertain world and making wise decisions so that you don't have to suffer any more than is necessary um uh yeah so that's that's you know like there would be even even if we even if reform is successful it would still be useful to develop to become better at problem solving um and i think there's great opportunity to become better i mean as a whole society hasn't even really tried to become a good problem solver like you know in a focused science driven way we've never really put our mind to this and we can put our mind to it and i think that would be tremendous improvements that we could make that's you know that's why i wrote the series is because i think we could do so much better than we're doing today which is not very um let's see anything else on this slide yes down in the lower right hand corner i have a kind of a long list of things that we need to do and i won't read it now but you know we basically have to remake everything from what we make how we make it how we use it where it goes how it's disposed of how we treat each other how we cooperate who we cooperate i mean just like deep transformation from across every aspect of human life and in human social life so uh you know it to me at least it's inconceivable that existing systems could even come close to the level of transformation that is actually not only needed but would be beneficial like it would be really beneficial if societies became good at problem solving and adaptation and it's pretty much inconceivable to me at least that existing systems and existing reform proposals and efforts could come close to what is actually needed and would be beneficial i think uh you know this is really great i'm just looking for a natural pause point for where we would want to be what do you think about doing 14 going through it taking our and then this r d program being the pickup point for our third session i know we're going slow you're going on some great boomerangs so it's awesome but let's walk through leaving somebody at the end of the second discussion what would an r d program be that would meet the scope and the type of these challenges three is going to be like you want to know what it is go there yeah and i think for for listeners daniel and i've been talking about you know there's a lot of material to unpack here so i think we're going to more than three videos uh but we'll figure out what the dates will be for them so that we can go through the whole thing uh you know carefully perfect um okay so uh we'll we'll end on on slide 14 then uh uh maybe to start i would just like to say that i've i'm the the series is a proposal for you know like a like a soft proposal for a new r d programming to know about development of new systems but other people might come up

with another proposal like you know mine maybe is just one of many possible ways to go about the r d program for new proposals and i would welcome i would hope even that it would be fantastic if there were you know 20 projects happening around of uh aimed at this same thing and that we and that each of these projects was cooperating with each other so that we can learn from each other and and move forward you know in the most efficient and useful most useful way so that's one of the reasons that i made desert list is uh like what would be if other if if there's going to be multiple r d programs or even if it's just my this just this one that i put out you know what what would be a good what has to be included in the r d program you know what which which what should be there so that we would know that it's a you know like a reasonably complete one and so that we could we could compare r d program so if i propose this one and someone else proposes another one we can compare say on number five uh number five develops new integrated systems that span all six of these major societal systems economic governance et cetera well if someone else proposes one that it's only focused on economic systems well there's a way to just distinguish between the two r d programs so that's uh that's that's part of my purpose here of laying this and then it's also also the purpose that you will see as you read the series that talk about each one of these so this is just in a sense it's a it's a index to the to the larger series and also one point um is that a scientist might say you need to do a b and c to really study you know the mosquitoes wing and then of course they measure a b and c so of course they solve the mosquitoes wing but actually you're laying out a protocol for evaluating policy across scales and then putting up almost like a retort to yourself at least initially that you believe could be held against this rubric like a report card and then again where a policy is better on point five or eleven or whatever it we can make that decision in a way or we can version the rubric so this isn't just setting the target or painting the target texas sharpshooter paint the target after you wrote the paper hypothesizing after the results are known harking this is like laying out where we do want it to go right exactly exactly and uh and i hope uh that we refined it together as a as a community refined this list so that we hone it better even um uh maybe there's some things i left out or some things i should have said or things i said i didn't need to say who knows but i'm sure it could be improved so uh we don't have much time and i won't go uh i won't go into detail on these but maybe i'll just uh just to give a flavor i'll just talk about a few scattering here um so obviously number one if you if you know you're talking about a r d program it might help to actually provide some information about the scale of the intended effect i mean are we thinking that this r d program is going to build uh new systems for a uh for one particular city

or for uh for you know for a nation or for you know like what scale is this whole thing going to apply who's going to where is it going to go if it's successful right what is the scale and what are the costs what are the risks how much money is this whole thing going to take you know and we'll talk about money we'll talk about in the next next talk uh based on second order science which i mentioned earlier already uh number three it has a theory of change like fine you you you you tested a new system you built it you get great computer simulation models great now what do we do with it how would you ever take that idea and make it real implement it in a real society somewhere how does it how does that work and and where is the blowback gonna be where is the pressure is gonna be against it what might go wrong so that should be in there uh theory of change um number five it develops integrate i claim that it's important that it develops integrated systems across all six of the major ones because if it doesn't you know if you if you build a governance system independent from an economic system and you're still using the old economic system it's you're it's not going to work very well because they're going to be at heads with each other not integrated um uh six describes the purpose of a system what in the world is it supposed to do how do we know it even worked what what do we measure fitness what what was it supposed to accomplish and how can we how can we evaluate it on what it did accomplish or what it's supposed to um uh how does cognition played like in this whole series as well focused on societal maybe some other ph maybe some other r d program doesn't even use the word cognition maybe they have an entirely different way to think about what a society is and how it does and what the purpose is so if so you know it'd be good to say so uh makes the work freely available to the public uh so that and i didn't we haven't talked about this but i i'm the the uh it's assumed here that whatever is developed would be developed in an open source way or would be freely available to the world this is not a proprietary project at all number eight it develops flexible options that the public can choose from if there is no one-size-fits-all solution i mean that wouldn't be very wise at all societies have to have flexibility in how they act how they what they implement what they choose so hopefully it would be a the program would develop a smorgasbord of ideas and possibilities that could be useful in different settings um number 11 addresses the cultural changes what kind of educational changes for example would be necessary to make this work uh there would be substantial ones actually because you build a new system for one thing you would have to explain or educate people on how the new system is even goes together and how it works and how what how would you know if it's not working you know all those kinds of questions um number 12 if new systems are implemented in

multiple locations so how do is that information shared between them how does that work is each independent or are they communicating or is this all part of a larger plan to have multiple tests going on at the same time or are those just independent things that come and go as they do uh um uh obviously number 13 develop systems that are fair and just and then you know you could that is reasonable uh and uh number 14 you know what are the expected impacts when you do simulations for this new system i was how does it look for energy use waste generation and climate change and biodiversity loss how you know what are we going to get if we um if we build these how is it going to go what is what is the likely outcome of these things so that's just a few ideas that a few topics that would be important to address in any r d program and i've tried my best to address them here and even in just three papers you know like some of these are addressed in just a few sentences or a few paragraphs somewhere in the papers because you know you would really need a book i think or a series a series of books to really do a thorough job on all of these well you know what i kind of saw almost when you have this list is like a github or a versioned open source repository where people can fork the rubrics they like they can add more challenges they can go into detail what do you mean by fair what do you mean that they're just well those are things that we actually in active inference we can get down to and that's not to say that we're going to be carving out the human it means that we're going to be designing for ourselves and for whoever's on board and just like the ship makers make it for those who are on the ship but the ship is part of society there's gonna be a ship with sales so i think they'll be such interesting conversations because really you could fractal into yeah seven what is the role of information i mean i went to a conference today on only a fragment of that so of course seven is going to be a big topic but it's a it's third order science well you did with laying out the second order criticism paradigm so it's quite interesting to read about and learn yeah you know speaking of uh speaking of a repository you know at the very least you need a language and a means of communication so that you can convey if i have an idea for new systems i need to convey that to the rest of the world in some way convey to you maybe you have your own idea of what a new system would look like so how do we convey that idea how do we convey a complicated idea to each other so that we know what each other is talking about and then we can start to do tests on them or figure out which one works better in what situations like just just just just to begin we just need a way to communicate a way to present the information so that we can understand what each other is saying just one kind of closing metaphor or at least thought that comes to mind is like let's say two people like a movie one person said i love the special effects the other one says the special

effects were super cheesy i thought the story the other person says i didn't even i don't like movies but i like the book so everyone can like the movie and the point is they're aligned on action they might even want to hang out together they might be part of the fan club they might have a product related to the movie but if the proj if the problem is framed as okay come to consensus on why you like this movie we cannot personalize the solution so everybody can have a different understanding of why they like the movie that's playing out in front of our eyes and if it's which part of this movie do you like or not like we're gonna be arguing a lot and if it's like which movies should we watch potentially that's a different question or how should we reduce uncertainty about the movie that we should watch there's a lot of ways to frame it that i would hope would be new yeah i think so i think so yeah um yeah maybe maybe we should leave it there and we'll pick up more uh next time and uh looking forward john thanks so much uh i'm gonna just re-upload this video in case there's any weirdness with the live stream so be high quality and just uh we'll be in touch about planning these upcoming sessions but we will know that we will meet next week on april 20th at 3 p.m pacific pdt for the third session of n of bleep okay sounds good see you later john peace okay bye bye